Audit of the SV efficiency proof, round 5: the LaTeX source after all corrections

Source audited: `sv_efficiency_proof.tex`, read in full, at the commit "Port the audit corrections ... into sv_efficiency_proof.tex". `sv_efficiency_proof.md` was checked alongside; every issue below is in both files. Earlier rounds: 1, 2, 3, 4. Not refereed.

Part A. Proof-check

A1. Scope

This round is a full read of the LaTeX file, which was written independently from the Markdown and then corrected. The focus is on interactions between corrections made in different rounds, especially the restated H3 (N4) against the local results, and the quadrature defect (R1) against older remarks.

A2. Findings

	where	issue	type		
	Theorem 4 parts 2–3 (proof, part 2, second bullet)	the local bound $	B_S'(e,v)	\le\max\{1,c_b\}\tnorm{e}_{\omega_a}\tnorm{v}_{\omega_a}$ is false for the Friedrichs-based entry of c_b (H3 as restated in N4)	**gap**
	Tool A2; Lemma G case 2, step 2	H^1_* denotes A2's test space (mean zero, or zero on $\partial\omega_a\cap\Gamma_D$), but §1 now uses $H^1_*(\omega)$ for $\{v=0\text{ on }\partial\omega\setminus\Gamma_N\}$	notation		
	Remark 5.4 ("vanish ... at the discrete solution"); Remark 5.3 (direct variant); Theorem 4, line "Together"	R1 was not carried into these places: at the discrete solution the defects equal the quadrature part, not zero	wording		
	§0, Tool A5 header, Remark 5.4, §7	A5 is still called "unverified / as far as I recall". Licht 2019's abstract (verified in round 2) confirms commuting smoothed projections that preserve partial boundary conditions; only the L^2 norm of the bound is unconfirmed	status		
	Lemma L	stated for $\ell\in\{\mathrm{Res}_A,\tilde S\}$, but its second inequality is applied to $B_A'(e,\cdot)$ in Theorem 4.1. That inequality is Tool A6 and holds for every $\ell\in(H^1_D)'$	statement		
	§0 ("bypassed when $\Gamma_N=\emptyset$ or $\sigma=0$"; the .md says "Lemma F", which does not exist); Remark 5.4 ("used only in Lemma H(ii)"); §7 ("for $\sigma>0$")	A5 is needed for $\sigma=0$ too (Remark 5.4 says so), $\Gamma_N=\emptyset$ means [CW08] suffices rather than A5 being bypassed, and A5 is now also used in Proposition 5.2	consistency		
	Lemma A, step 2	"$m_K=\sigma^{-1/2}$" holds only for $\theta_K\le1/\pi$. The proof only needs $m_K\le\sigma^{-1/2}=h_K\varepsilon^{-1/2}\theta_K$, which always holds (noted in round 1, never fixed)	wording		

Everything else is re-checked and correct. That covers Theorems 1, 2 and 6, Corollaries 3 and 3', Lemmas A, B, G, H, D, L and V, Proposition 5, and the §1 quadrature defect.

The `.tex` matches the `.md` in content: section and result numbering (Theorems 1, 2, 4, 6; Corollaries 3, 3′; Proposition 5), all statements and all numbers. It compiles cleanly.

A3. T5 in detail

- **Where.** In Theorem 4 part 2, $|\mathrm{Res}|_{,a}\le\max\{1,c_b\}\tnorm{e}_{\omega_a}+|B_A'(e,\cdot)|_{,a}$ uses

$$|((c-\tfrac12\mathrm{div}\,b)e,v)_{\omega_a}|\le|c-\tfrac12\mathrm{div}\,b|_\infty|e|_{\omega_a}|v|_{\omega_a}\le c_b\,\tnorm{e}_{\omega_a}\tnorm{v}_{\omega_a}.$$

- **Which entry works.** The last step holds for the σ-based entry, via $|e|_{\omega_a}\le\sigma^{-1/2}\tnorm{e}_{\omega_a}$. For the Friedrichs-based entry it fails: $|e|_{\omega_a}$ is not controlled by $\varepsilon^{1/2}|\nabla e|_{\omega_a}$ on a patch, because e does not vanish on $\partial\omega_a$. C_F is a global constant on $H^1_D(\Omega)$.
- **Status.** The gap was introduced by N4, which widened H3. Before N4, H3 allowed only the σ-based c_b (with $c=\frac12\mathrm{div}\,b$ when $\sigma=0$), and Theorem 4 was correct.
- **Fix,** either of:
  1. **Restrict.** In Theorem 4 parts 2–3, take c_b to be the σ-based entry. That is, assume $\sigma>0$, or $c=\frac12\mathrm{div}\,b$ when $\sigma=0$. The global results (Corollaries 3 and 3', Proposition 5) keep the Friedrichs-based entry.
  2. **Augment the local measure.** For $v\in H^1_*(\omega_a)$, the local Friedrichs inequality $|v|_{\omega_a}\le C_Ph_a|\nabla v|_{\omega_a}$ holds (v vanishes on $\partial\omega_a\setminus\Gamma_N$, which has positive measure). Hence $|((c-\frac12\mathrm{div}\,b)e,v)_{\omega_a}|\le C|c-\frac12\mathrm{div}\,b|_\infty h_a\varepsilon^{-1/2}|e|_{\omega_a}\tnorm{v}_{\omega_a}$. Then add $\sum_a(|c-\frac12\mathrm{div}\,b|_\infty h_a\varepsilon^{-1/2}|e|_{\omega_a})^2$ to $E_{\rm loc}^2$ when σ = 0. It is a weighted L^2 error term, of higher order where $h_a\ll\varepsilon^{1/2}$.

Option 1 is the minimal fix. Option 2 keeps Theorem 4 for every case the code covers.

- **The code is unaffected.** `eta_Eloc` is the upper bound of part 1, which holds for both entries.

A4. Fixes for the minor items

- **T1.** Rename A2's space $H^1_\dagger(\omega_a)$ in Tool A2 and Lemma G case 2.
- **T2.** In Remark 5.4: "which reduce to the quadrature part ($\le C\,\mathrm{osc}_q$) for $\ell=\mathrm{Res}_A$ at the discrete solution". Similarly in Remark 5.3 ("$\ell(\psi_a)=S(u_h;u_h,\psi_a)$ up to $\delta^{\rm quad}$") and in Theorem 4's "Together" line ("$E\lesssim E_{\rm loc}+\eta_{\rm def,S}+C\,\mathrm{osc}_q$").
- **T3.** Write "abstract-level: preserves partial boundary conditions, commutes, uniformly bounded; the L^2 norm of the bound to be confirmed in the full text".
- **T4.** State the second inequality of Lemma L for every $\ell\in(H^1_D)'$.
- **T6.**
  - In §0: "used in Lemma H(ii) and Proposition 5.2; for $\Gamma_N=\emptyset$ it is [CW08], for mixed boundary conditions [Licht19]".
  - In §7: drop "for σ > 0".
  - In the .md: replace "Lemma F" by "Lemma H(ii)".
- **T8.** Write "$m_K\le\sigma^{-1/2}=(h_K/\varepsilon^{1/2})\theta_K$".

Status of the fixes (18 Oct 2026; applied to both `.tex` and `.md`)

- **T5:** fixed with both options.
  - Theorem 4 parts 2–3 are stated with the σ-based $c_{b,\sigma}$ (σ > 0, or $c=\frac12\mathrm{div}\,b$). The proof of part 2 now says why the Friedrichs-based entry is not admissible locally.
  - New Remark 9.3 treats σ = 0 with $c\ne\frac12\mathrm{div}\,b$ through the local Friedrichs inequality on $H^1_*(\omega_a)$. Part 2 holds with $E_{\rm loc}^2$ augmented by $\sum_a\mu_a^2$, where $\mu_a=C_P|c-\frac12\mathrm{div}\,b|_{\infty,\omega_a}h_a\varepsilon^{-1/2}|e|_{\omega_a}$. Part 3 holds with the same augmentation and the constant $C\kappa$, since its first step is global.
- **T1:** A2's test space is renamed $H^1_\dagger(\omega_a)$ (Tool A2, Lemma G case 2).

- **T2:** Remark 5.3, Remark 5.4 and the "Together" line of Theorem 4 now carry the quadrature defect.
- **T3:** A5 is described at abstract level in §0, in the Tool A5 header, in Remark 5.4 and in §7.
- **T4:** the second inequality of Lemma L is stated for every $\ell\in(H^1_D)'$.
- **T6:** §0 now says A5 is used in Lemma H(ii) and Proposition 5.2, for σ > 0 and σ = 0 alike ([CW08] for $\Gamma_N=\emptyset$). Remark 5.4 and §7 are corrected. The .md's "Lemma F" is gone.
- **T8:** Lemma A step 2 reads $m_K\le\sigma^{-1/2}$.
- The `.tex` compiles cleanly (22 pages), and the result numbering is unchanged.

A5. Optimality

- **T5 refines the hypotheses-used list.** The Friedrichs-based c_b is admissible in the global results only. The local results (Theorem 4.2–4.3) need the σ-based one or the extra local L^2 term.
- **Otherwise unchanged from round 4.** The c_b-free track (cellwise σ, adapting AV14) would also remove T5, since it puts the reaction weight into the norm. That is one more reason to prefer it.

Part B. Literature scan (round 5)

B1. Answer

There is no change in novelty or track. The round-4 open item is resolved at abstract level: EARM and Li 2026 are compatible.

B2. EARM against Li 2026

- **EARM** (He, arXiv 2503.03653 and 2601.01585) proves **robust efficiency** of an indicator based on the difference between the numerical flux and the equilibrated recovered flux. The efficiency constant is independent of the coefficient jump, for DG and conforming discretisations.
- **Li 2026** (arXiv 2609.27817) proves that for conforming P1 at a non-quasi-monotone checkerboard vertex, two instances that are identical on any fixed number of local mesh layers have energy errors that differ by a factor of order $M^{1/4}$. Hence no local estimator has $C_{\rm rel}C_{\rm eff}$ independent of the contrast. This covers residual, equilibrated and recovery estimators.
- **Reconciliation.** Robust efficiency alone does not contradict Li. The obstruction is the product with reliability. Either EARM's reliability constant, or the error measure in which it is two-sided, must carry the contrast in the checkerboard case. Confirm in the full texts which it is.
- **For §11 the statement stays as written:** Li's result shows that a two-sided, contrast-robust bound without quasi-monotonicity is impossible; EARM can be cited as robust efficiency.

B3. Unchanged

- Licht 2019 is at abstract level.
- AV11 and AV14 are verified.
- Nothing new on equilibrated estimators for algebraic stabilisations, 2025–2026.

B4. Open items, ranked by novelty-to-effort

1. **T5:** restrict, or add the local L^2 term, in Theorem 4. Low effort; needed for correctness.
2. **T1–T4, T6, T8:** text only.
3. **The c_b-free track** (cellwise σ). Medium effort; it removes the c_b dependence and T5's restriction.
4. **A5:** read Licht 2019, Theorem 1. **EARM/Li:** read the full texts to confirm the reconciliation.
5. **The stopped iterate against the discrete solution.** Open, high risk.

B5. References (new in round 5)

- C. He, Equilibrated averaging residual method: a general approach to conservative flux recovery, arXiv 2503.03653; A unified equilibrated flux recovery framework with robust a posteriori error estimation, arXiv 2601.01585. [abstracts]
- Y. Li, arXiv 2609.27817. [abstract; the author attribution is from round 1's search]

- Cai–Zhang higher-order conforming equilibration under quasi-monotonicity of a projected coefficient: Emory. [abstract]